LaTeX Training Workshop for Researcher and Professional

Hands On Approach

By

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Bachelor's Thesis

Submitted in the partial fulfillment of requirements for the degree of Bachelor of Science in Computer Science with specialization in Software Engineering

Department of Computer Science
Faculty of Science and Information Technology
Sukkur IBA University

Defended publicly on 13 APRIL 2026

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DECLARATION

I/We, hereby declare that this thesis has been composed solely by myself/us, during the period of my Bachelor's study. I am aware of and understand the university's & HEC policy on plagiarism and I certify that this thesis is my own work, except where indicated by reference, and the work presented in it has not been submitted in support of another degree or qualification from this or any other university or institute of learning. If a violation of the university & HEC policy on research has occurred in this thesis then I shall be liable for punishable action under the rules of the university & HEC.

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Science is a wonderful thing
if one does not have to earn one's living at it.

Albert Einstein

SUMMARY

This thesis presents a review of the article entitled “A dynamic model for HIV infection”. The model of HIV infection with and without treatment is considered and revisited to monitor disease progression. In this thesis, three HIV models are considered and analyzed numerically. The antiretroviral therapy in the model of HIV dynamics is considered, which gives rise to two input control parameters:

- Reverse Transcriptase Inhibitors (RTIs)
- Protease Inhibitors (PIs)

The RTIs control infection of CD4+ cells and PIs prevent production of new virions (free virus).

The advantage of this model is that it is easy to control the nonlinear terms representing the infection of healthy CD4+ cells by HIV virus and the production of new virions with different drug dosages of RTIs and PIs, respectively. If the treatment is terminated after a certain time, then infected CD4+ cells and virus load remain low for several days, even up to 150 days.

In order to further simplify the antiretroviral therapy model of HIV dynamics, the effects of RTIs and PIs are combined as a single input parameter in the model. The advantage of this model is that it controls only production of new virions, but the drawback of this model is that if the treatment is terminated after a certain time then viral load rises up much faster than that of antiretroviral therapy model of HIV dynamics.

The governing nonlinear dynamical system of HIV dynamics under treatment is linearized around its critical/equilibrium point to study the asymptotic behavior of the dynamical system. The analytical solution is computed around the equilibrium point and stability of the nonlinear system around the critical point is tested and the results of linearized system are presented through various graphs. The Hartman-Grobman theorem is used to guarantee that nonlinear system around the hyperbolic point behaves same as the linearized system. Moreover, the HIV dynamical systems under consideration are solved and compared with Runge-Kutta method of order 4. The benefit of using Runge-Kutta method of order 4 is that global approximated solution is found subject to given initial conditions.

Acknowledgements

Express gratitude to your supervisor, co-supervisor, department, peers, or any funding organizations who supported the project.

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1

INTRODUCTION

Albert EINSTEIN

Nature and nature's laws lay hid in the night;
God said 'Let Newton be!' and all was light.

Alexander Pope

It did not last: the devil shouting .
Let Einstein be!' restore the status quo.

Sir John Collings Squire

This document is intended to be both an example of the SIBAU dissertation template for LaTeX, as well as a short introduction to its use. It is not intended to be a general introduction to LaTeX itself,[1] and we will assume the reader to be familiar with the basics of creating and compiling documents.

Instructions on how to use this template under Windows and Linux, and which LaTeX packages are required, can be found in `README.txt`. [3]

1.1. Document Structure

Since a dissertation is a substantial document, it is convenient to break it up into smaller pieces. In this template we therefore give every chapter its own file. The chapters (and appendices) are gathered together in `dissertation.tex`, which is the master file describing the overall structure of the document. `dissertation.tex` starts with the line

This is Section 1.1 of Chapter 1

```
\documentclass{dissertation}
```

which loads the dissertation template. The template is based on the LaTeX `book` document class and stored in `dissertation.cls`. The document class accepts several comma-separated options. By default, hyperlinks are shown in cyan, which is convenient when reading the dissertation on a computer, but can be expensive when printing. They can be turned black with the `print` option. This will also turn the headers dark gray instead of cyan. Moreover, it will add a 3 mm bleed around the page including crop marks. This will help the printer with the thumb indices, since they run right up to the page borders. Finally, the `nativefonts` option can be used to override the automatic font selection (see below).

A dissertation is a big document, which makes it easy to miss warnings about the layout in the LaTeX output. In order to locate problem areas, add the `draft` option to the `\documentclass` line. This will display a vertical bar in the margins next to the paragraphs that require attention.

The contents of the dissertation are included between the `\begin{document}` and `\end{document}` commands, and split into three parts by

1. `\frontmatter`, which uses Roman numerals for the page numbers and is used for the title page and the table of contents;

2. `\mainmatter`, which uses Arabic numerals for the page numbers and is the style for the chapters;

3. `\appendix`, which uses letters for the chapter numbers, starting with 'A'.

The title page is defined in `title.tex` in the `title` folder and included verbatim with `\include{title/title}`,[2] (see below). Additionally, it is possible to include a preface, containing, for example, the acknowledgements. An example can be found in `preface.tex`. The table of contents is generated automatically with the `\tableofcontents` command.

[1] We recommend `http://en.wikibooks.org/wiki/LaTeX` as a reference and a starting point for new users.
[2] Note that it is not necessary to specify the file extension.

Chapters are included after `\mainmatter` and appendices after `\appendix`. For example, `\include{chapter-1/chapter-1}` includes `chapter-1.tex`, which contains this introduction.

1.2. TITLE PAGE

THE title pages are defined in `title/title.tex`, which you will have to modify according to your needs. Note that these pages are subject to the requirements of the *promotieregelement* and cannot be changed at will.

Since the thesis title and name of the author appear several times throughout the document (on the title page, but also in, *e.g.*, the preface and cv), special commands are provided so they only have to be specified once. The title (and optional subtitle) can be specified with

```
\title[Optional subtitle]{Title}
```

The name of the author is specified with

```
\author{First name}{Last name}
```

Note that the first and last name are separate arguments, since they may be printed in different font shapes. The `\title` and `\author` commands also ensure that the title and author appear in the metadata of the final PDF.

See `title/title.tex` for detailed documentation on the comment and layout of the title pages. Logos of institutes that have contributed financially to the dissertation may be included on reverse side of the title page. A few example logos can be found in the `title/logos` folder.

1.3. CHAPTERS

EACH chapter has its own file. For example, the LaTeX source of this chapter can be found in `chapter-1.tex`. A chapter starts with the command

```
\chapter{Chapter title}
```

This starts a new page, prints the chapter number and title and adds a link in the table of contents. If the title is very long, it may be desirable to use a shorter version in the page headers and the table of contents. This can be achieved by specifying the short title in brackets:

```
\chapter[Short title]{Very long title with many words which could
not possibly fit on one line}
```

Unnumbered chapters, such as the preface, can be created with `\chapter*{Chapter title}`. Such a chapter will not show up in the table of contents or in the page header. To create a table of contents entry anyway, add

```
\addcontentsline{toc}{chapter}{Chapter title}
```

after the \chapter command. To print the chapter title in the page header, add

```
\setheader{Chapter title}
```

If (parts of) the chapter have already been published elsewhere, it is customary to add a reference. This can be done with the special unnumbered footnote command \blfootnote. For example,

```
\blfootnote{Parts of this chapter have been published in Annalen
der Physik \textbf{324}, 289 (1906) \cite {Einstein1906}.}
```

generates the footnote at the beginning of this chapter. Because this footnote is unnumbered, the hyperref package may throw a warning, which safely be ignored.

If multiple people have contributed significantly to this chapter, they can be lister with the \authors command. This can be followed by a quotation using \epigraph as shown above. Finally, it is customary for a dissertation to include an abstract for every chapter (except perhaps the introduction). This can be accomplished with the abstract environment. The abstract should be followed by \newpage to start the chapter text on a new page.

In a dissertation, each chapter has its own list of references. These can be generated with the special command \references{dissertation} from dissertation.bib at the end of the chapter. Note that this means that you need to run a command like bibtex chapter-1/chapter-1 for each chapter. The bibliography style is specified in dissertation.bst, which is a modified version of apsrev4-1.bst (from REVTeX) designed to also display the titles of referenced articles. The template will automatically generate clickable hyperlinks if a URL or DOI (digital object identifier) is present for the reference. Although it is possible to manage the bibliography by hand, we recommend using EndNote (available from Blackboard) or JabRef (available from http://jabref.sourceforge.net/).

Chapters are subdivided into sections, subsections, subsubsections, and, optionally, paragraphs and subparagraphs. All can have a title, but only sections and subsections are numbered. As with chapters, the numbering can be turned off by using \section*{...} instead of \section{...}, and similarly for the subsection.

1.4. \SECTION{...}

1.4.1. \SUBSECTION{...}

\SUBSUBSECTION{...}

\paragraph{...} Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

1.5. FONTS AND COLORS

THE fonts used by this template depend on which version of LaTeX you use. Regular LaTeX, *i.e.*, if you compile your document with with `latex`, `pslatex` or `pdflatex`, will use Utopia for text, Fourier for math and Latin Modern for sans-serif and monospaced text. However, if you want to adhere to the SIBAU house style, you will need to use XeLaTeX, as it supports TrueType and OpenType fonts. Compiling with `xelatex` will use Bookman Old Style for titles, Tahoma for text, Courier New for monospace and Cambria for math. If you want to use XeLaTeX, but do not want to use the SIBAU house style fonts, you can add the `nativefonts` option to the document class.

This template supports the use of drop caps, a large colored initial at the beginning of a chapter or section, via the `\dropcap` command:

```
\dropcap{L}{orem} ipsum...
```

The first argument is the capital that will be printed on two lines (in the title color), and the second argument is the rest of the word. Depending on the font, the latter may be printed in small caps.

The corporate colors of the SIBAU are dark blue, black and white, available, respectively, via `\color{sibau-cyan}`, `\color{sibau-black}` (which differs slightly from the default `black`) and `\color{sibau-white}`. Apart from these three, the house style defines the basic colors

- `sibau-sea-green`,
- `sibau-green`,
- `sibau-dark-blue`,
- `sibau-purple`,
- `sibau-turquoise` and
- `sibau-sky-blue`,

as well as the accent colors

- `sibau-lavendel`,
- `sibau-orange`,
- `sibau-warm-purple`,
- `sibau-fuchsia`,
- `sibau-bright-green` and
- `sibau-yellow`.

1.6. CITATION GUIDELINES

This document explains how to properly manage citations and references using this template, which is configured to use the modern biblatex package.

1. **Quick Start: How to Add a Citation**
   Add an entry to the dissertation.bib file.

2. Cite it in your text using the \cite {key} command.

3. Compile your document with the correct sequence: XeLaTeX ⟶ Biber ⟶ XeLaTeX ⟶ XeLaTeX.

1.6.1. THE BIBLIOGRAPHY FILE (DISSERTATION.BIB)

All your references are stored in the dissertation.bib file in the BibTeX format. You can manage this file manually or use a reference manager like JabRef or Zotero.

1.7. CITING IN YOUR TEXT

Use the \cite{ } command with the appropriate key.

Single citation: \cite{Moysis2016}
Multiple citations: \cite{Moysis2016, Nagle2011}
Cite with page number (pretext): As shown on page 123 of \cite[123]{Nagle2011}
Cite with note (posttext): \cite[See especially Chapter 5]{Nagle2011}

EXAMPLE

Numerical methods are used to approximate solutions when an exact one is unavailable [1]. In this work, the Runge-Kutta method (RK4) is applied to the HIV dynamical model presented by [4]. This model, which considers antiretroviral therapy, gives rise to two key control parameters [3, 2].

2

Literature Review

Numerical methods are used to approximate the solution of equations when exact solutions cannot be determined. Numerical methods help find the global behavior of ordinary differential equations. The HIV dynamical models discussed in Chapter 1 are nonlinear, and to approximate their solutions, the Runge-Kutta method of order 4 is used [1].

Table 2.1: Values of the constant parameters used in the system.

Description	**Symbol**	**Value**
Time	t	days
Death rate of uninfected T cells	d	0.02 per day
Rate of virions produced	k	100 counts per cell
Production rate of uninfected T cells	s	100 mm^{-3} per day
Ineffective rate of virions	β	2.4×10^{-5} mm^{-3} per day
Death rate of virus	m_1	2.4 per day
Death rate of infected T cells	m_2	0.24 per day

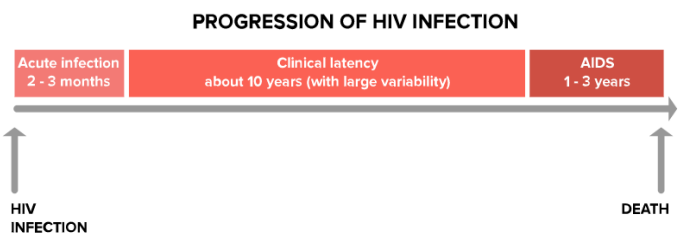

Figure 2.1: Simulations of the dynamics model.

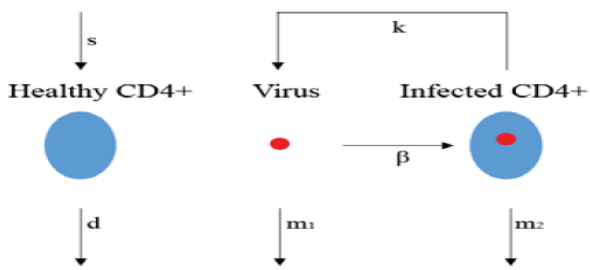

Figure 2.2: Simulations of the dynamics model.

3

Research Methodology

Explain the proposed method, model, algorithms, or frameworks used. Describe system architecture, design diagrams, and workflow.

4

Implementation

Explain how the proposed methodology was applied, tools and technologies used, datasets, software, or hardware setup.

5

Results and Discussion

Present results using tables, graphs, and figures. Compare with existing solutions. Discuss significance and limitations.

6

Conclusion and Future Work

Summarize achievements, reflect on contributions, and propose future directions for improvement or extension.

BIBLIOGRAPHY

[1] Richard L. Burden and J. Douglas Faires. *Numerical Analysis*. 9th ed. Brooks Cole, 2010. ISBN: 0538733519. URL: https://www.cengage.com/.

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[4] Lazaros Moysis, Ioannis Kafetzis, and Marios Politis. "A dynamic model for HIV infection". In: *Conference Paper* (2016). Presented at the 24th Mediterranean Conference on Control and Automation.